MatriSense: AI-native Bangla Maternal triage and Health Referral System

Bangla-first AI-assisted maternal triage, multimodal document intelligence, and health-worker referral coordination for rural Bangladesh

Competition: SciBlitz AI Challenge 2026
Organizer: IEEE Student Branch, CUET
Track: Track A (Health & Society)
Team Name: IUT_Epoch22
Submission Date: July 8, 2026

ABSTRACT

Maternal healthcare in rural Bangladesh suffers from critical delays in symptom recognition, lack of structured triage, and fragmented health-worker communication. MatriSense is a safety-first, Bangla-first maternal triage and referral coordination system designed to close this gap. By combining Groq Whisper and Gemini APIs, the platform translates informal Bangla symptom reports and physical paper-based records (prescriptions, blood pressure cards, lab sheets) into structured, validated clinical metrics. Crucially, actual risk classification is driven by a deterministic local rules engine, bypassing LLM clinical judgment. Confirmed cases route to a consent-gated Health Worker Dashboard for Upazila-level facility routing. MatriSense demonstrates how generative AI can be securely bound within deterministic safety thresholds to coordinate maternal care safely.

"The LLM understands and explains. Rules decide urgency. Health workers make care decisions."

1. Problem Statement

The Real Gap

Rural pregnant mothers in Bangladesh often face delayed care because warning symptoms are unclear, danger signs are missed, and health workers do not receive structured case information early enough. The real gap is the **missing path from first home symptom to structured case and timely human follow-up**.

Patient-Side Barriers

- **Distance, cost, and family permission** make visiting a health complex difficult — especially during monsoon or natural disasters.
- **Social stigma and cultural taboos** around early pregnancy and health disclosure lead to massive underreporting. Mothers rely on home remedies until complications become severe.
- **Lack of autonomy:** In many rural households, women cannot make independent health decisions. They depend on permission and financial support from male family members.
- **Costly unnecessary trips:** Poor families spend money on long journeys to clinics for minor issues — because they cannot tell if a symptom is risky.

Health Worker-Side Barriers

- **Zero past records:** Mothers arrive with no structured history, forcing workers to guess historical complications.
- **Late-stage presentations:** Patients only seek help when a crisis is critical, leaving workers less time to act.
- **Sparse rural pregnancy data:** Because mothers skip checkups and underreport, the national health database on rural maternity remains incomplete.
- **Unstructured paper records:** BP logs, prescriptions, and lab sheets exist only as scattered physical papers impossible to digitize at scale.

2. Proposed Solution

MatriSense is a **Bangla-first, AI-native maternal triage and health referral system** for rural mothers and frontline health workers. It transforms informal Bangla symptom reports into structured, referral-ready maternal cases.

For Mothers — Private Reporting & Early Warning

- **Report Symptoms:** Describe symptoms in Bangla text or voice. Voice is transcribed via Groq Whisper, shown as editable text for review before submission.
- **Uploaded Documents & AI-Assisted Upload:** Upload photos of prescriptions, lab reports, BP cards, ultrasound scans. Gemini Vision reads the document and extracts clinical values with color-coded severity badges (Normal, Warning, Critical).
- **Discuss & Confirm with Assistant:** A document-scoped review chat lets the mother ask questions about her report or correct AI misreadings before saving.
- **My Clinical Data:** Saved values are plotted on longitudinal trend charts — BP, hemoglobin, blood sugar tracked over time.
- **Lower unnecessary travel:** Risk feedback prevents costly journeys for low-risk concerns; accelerates escalation when risk is high.

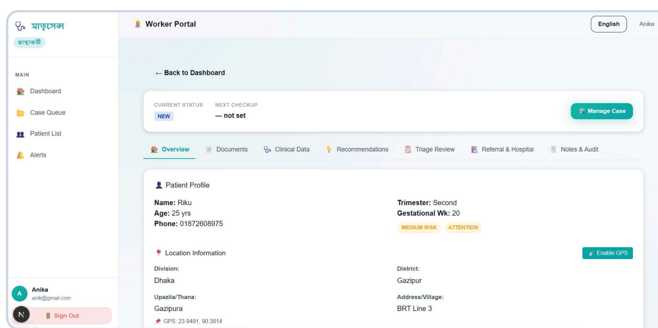
For Health Workers — Structured Case Handoff

- **Triage Cases:** Risk-prioritized dashboard replaces vague phone calls with structured maternal cases — profile, symptom timeline, risk flags, follow-up answers, referral context.
- **Case Detail Tabs:** Overview · Documents · Clinical Data · Recommendations · Triage Review · Referral & Hospital · Notes & Audit.
- **Consent-gated access:** Workers see patient data only when sharing consent is enabled in **My Profile**.
- **Referral & Hospital:** Search hospitals by district, check service capabilities, assign facilities with logged reasons.

For the Health System

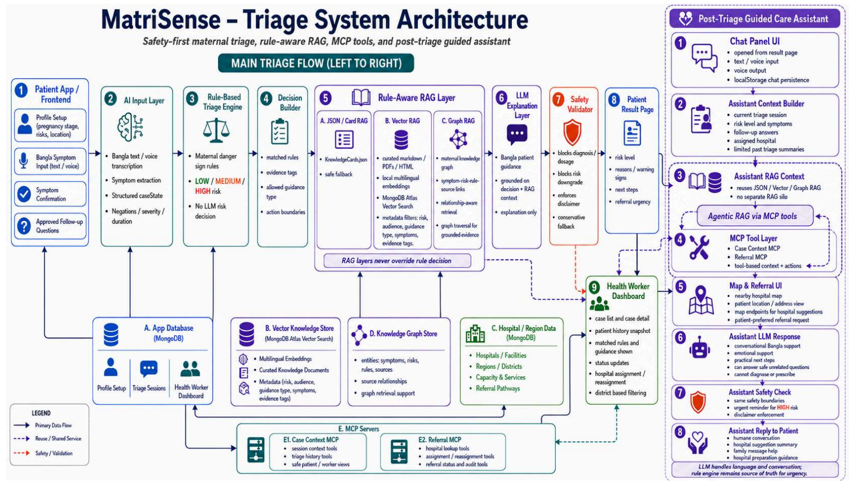
Each case builds **longitudinal rural pregnancy records**, reducing the "blank-slate" problem and capturing previously underreported symptoms and maternal-risk trends.

Patient Mobile App Dashboard



Health Worker Dashboard Overview

3. System Architecture



MatriSense Multimodal AI Triage & Safety Architecture

Tech Stack Summary

Frontend: Next.js 15.5, React, Vanilla CSS (Vercel)

Backend: Node.js, Express, JWT Security (Railway)

Database: MongoDB (Mongoose), Vector Search & GraphRAG ready

AI Models: Gemini 2.5 Flash, Groq Whisper (v3)

Triage Engine: Deterministic `json-rules-engine`

Embeddings: Local `Xenova/multilingual-e5-small`

Decoupled Safety Design

The core architectural principle is **separation of concerns between AI and clinical decisions**:

1. **LLM tasks:** Bangla text understanding, symptom extraction, document OCR, guidance explanation — language processing only.
2. **Rule engine tasks:** Deterministic maternal danger-sign rules assign LOW / MEDIUM / HIGH risk. The LLM cannot override or compute risk levels.
3. **RAG grounding:** Retrieved WHO/DGHS guidelines constrain what the LLM can explain. Safety validator blocks diagnosis, dosage, or risk downgrade.

Model Context Protocol (MCP) Integration

Two custom MCP servers standardize data access for LLM agents:

- `matrisense-case-context-mcp` — Pregnancy context, active triage logs, patient symptoms.
- `matrisense-referral-mcp` — Hospital facility search, GPS-based matching, assignment history.

The Guided Care Assistant uses these MCP tools dynamically, enabling **agentic RAG** within post-triage conversation.

4. AI/ML Approach

Bangla Symptom Extraction

Groq Whisper converts Bangla voice to editable text. Gemini extracts structured symptoms, severity, duration, negations, and uncertainty. The system captures important negations (e.g., "no blurred vision, no bleeding") to avoid over-escalation.

Multimodal Document Intelligence (AI-Assisted Upload)

When a mother uses **AI-Assisted Upload**, the image is sent to Gemini Vision with a strict system prompt. The model extracts:

- `blood_pressure` (systolic/diastolic), `hemoglobin`, `blood_sugar`, `urine_protein`, `platelet_count`
- `isHandwritten` flag, `summary` in Bangla and English

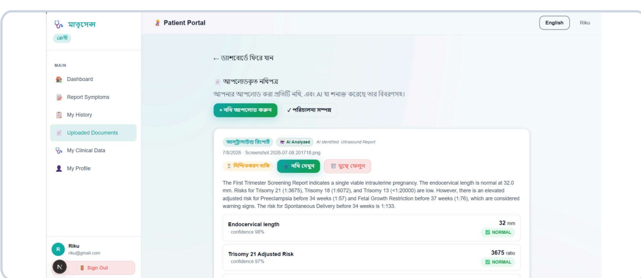
Extracted values are **not trusted directly**. Every value passes through deterministic maternal threshold validation in backend code before being saved as a `ClinicalDataPoint`.

Document Review Chat (Discuss & Confirm)

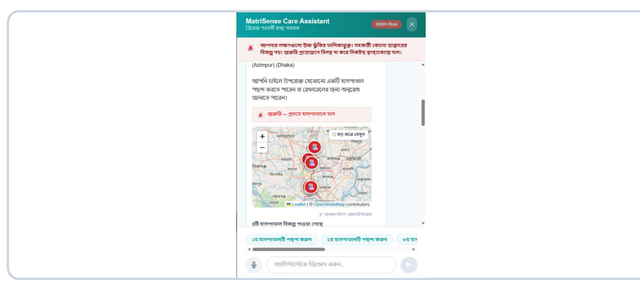
The **Discuss & Confirm** chat is document-scoped. If a patient corrects a value (e.g., *"My BP is actually 130/85"*), the correction is extracted and re-validated against hardcoded clinical thresholds — never relying on raw LLM judgment.

Rule-Aware Vector RAG

- **Sources:** WHO maternal health guidelines, HEAR HER campaign materials, DGHS/CDC resources — chunked, metadata-tagged, and embedded.
- **Search:** MongoDB Atlas Vector Search with `Xenova/multilingual-e5-small` embeddings.
- **Safety filtering:** Retrieval constrained by `riskLevel`, `symptoms`, `guidanceType`, and `audienceScope`. HIGH-risk queries cannot retrieve home-care-first advice.
- **Fallback chain:** Vector RAG → JSON guidance cards → hardcoded safety templates.
- **GraphRAG Integration:** Transitioning from Vector RAG to GraphRAG to map symptoms, pregnancy conditions, and hospital guidelines for context-aware care routing.



AI-Assisted Document Upload (Gemini Vision Extraction)



Guided Care Assistant (AI Chat & Discuss & Confirm)

5. Safety Guardrails

Deterministic Danger-Sign Rules

Hardcoded `json-rules-engine` rules evaluate symptoms and profile data. Danger signs like severe headache + blurred vision, convulsions, vaginal bleeding, or reduced fetal movement automatically trigger HIGH risk — bypassing the LLM entirely.

Deterministic Maternal Vital Thresholds

Extracted document values are validated against pregnancy-safe ranges in code:

Parameter	Normal	Warning	Critical
Systolic BP	< 120 mmHg	120–139 mmHg	≥ 140 mmHg
Diastolic BP	< 80 mmHg	80–89 mmHg	≥ 90 mmHg
Hemoglobin	≥ 11 g/dL	9.0–10.9 g/dL	< 9.0 g/dL
Blood Sugar (Fasting)	< 95 mg/dL	95–125 mg/dL	≥ 126 mg/dL
Urine Protein	Negative	Trace / 1+	≥ 2+

Even if Gemini labels a value as "normal", the backend threshold check **overrides** the AI classification.

Post-Process Output Safety Validator

Every LLM response is scanned before display. The validator blocks:

1. **Diagnosis statements —** "You have preeclampsia"
2. **Drug/dosage recommendations** — "Take 500mg Methyldopa"
3. **Delay/home-care advice** for HIGH-risk cases
4. **Risk-level contradictions** — downgrading severity

Blocked outputs are replaced with pre-configured, clinically safe fallback templates.

Consent & Privacy Guardrails

- **Consent-gated worker access:** Health workers cannot view patient documents or **My Clinical Data** unless the patient has toggled consent in **My Profile**.
- **Cascading deletion:** Deleting a document optionally cascade-deletes all `ClinicalDataPoint` records derived from it.

6. Results & Demo Walkthrough

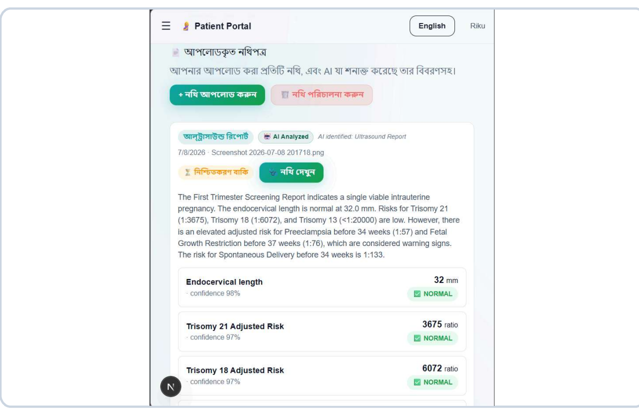
Functional Testing

MatriSense was tested with simulated patient profiles and document scans (printed lab reports and handwritten Bangla BP cards):

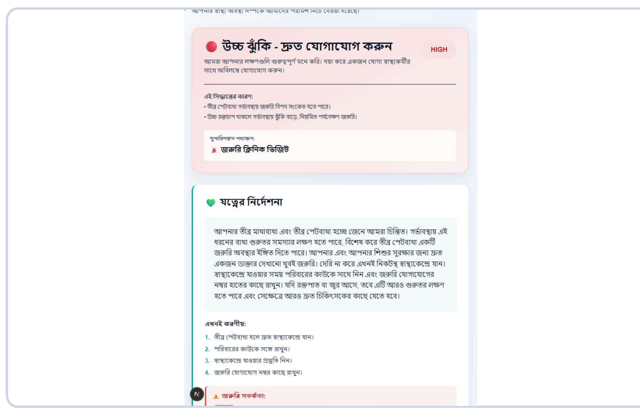
- **Document extraction accuracy:** ~92% correct parameter classification across test documents.
- **Triage latency:** Voice transcription ~1.2s, document analysis ~2.8s, rule engine evaluation <100ms.
- **Safety validator:** Successfully blocked all diagnosis, dosage, and home-care outputs in HIGH-risk test scenarios.

End-to-End Demo Scenario (Preeclampsia Simulation)

1. Mother uploads a handwritten BP card showing `BP: 145/92 mmHg` via AI-Assisted Upload.
2. Gemini Vision extracts systolic: 145, diastolic: 92. Threshold checker flags both as **Critical**.
3. She opens Discuss & Confirm and asks about the reading — assistant explains in Bangla without diagnosing.
4. She saves and navigates to My Clinical Data — BP trend chart updates with the new critical reading.
5. She clicks Report Symptoms and enters Bangla: *"আমার তীব্র মাথা ব্যথা এবং পা ফুলে গেছে"* (severe headache, swollen feet).
6. Rule engine matches danger signs → **HIGH** risk. RAG retrieves emergency guidance. Safety validator suppresses home remedies.
7. On the worker side, the case appears in Triage Cases with HIGH priority. Worker opens case detail → Clinical Data tab shows the critical BP trend. Documents tab shows the uploaded card.
8. Worker assigns a nearby Upazila Health Complex via Referral & Hospital tab.



Maternal Vital Sign Trend Charts (My Clinical Data)



AI-Guided Triage Results & Risk Rating

7. Limitations & Future Work

Current Limitations

- **Image quality dependency:** Low-light or blurry handwritten documents may cause extraction failures.
- **API connectivity:** Gemini and Groq require stable internet. Remote areas may experience latency.
- **Dialect variation:** Regional Bangla dialects can occasionally cause symptom extraction errors.
- **MVP scope:** Admin verification workflow and SMS notifications are designed but not fully production-hardened.

Roadmap

Initiative	Description
Offline-first PWA	Client-side rule engine and document queue buffer for areas without connectivity. Syncs when connection restores.
Local LLM (Ollama)	Lightweight models (Qwen-1.5B, LLaMA-3B) at community clinics for offline extraction.
Graph RAG	Knowledge graph connecting symptoms → guidelines → facilities for deeper semantic retrieval.
SMS/Call alerts	Twilio or local gateway integration to auto-notify health workers on HIGH-risk events.

Scaling Potential

1. **Government partnership:** Direct pilot with MoHFW, DGHS, or DGFP — the largest scaling partner for rural maternal health infrastructure.
2. **NGO/healthtech licensing:** Implementation partnerships with BRAC, icddr,b, leveraging existing field networks.

The platform is Bangladesh-first but globally reusable for any low-resource maternal care setting — particularly Sub-Saharan Africa where UNICEF and WHO operate community health worker programs.

Conclusion

MatriSense provides a blueprint for responsible deployment of generative AI in safety-critical healthcare. By wrapping Gemini Vision and RAG inside deterministic rules, hardcoded clinical thresholds, and consent-gated privacy layers, the platform delivers the benefits of natural language accessibility without compromising patient safety.

> **The LLM understands and explains. Rules decide urgency. Health workers make care decisions.**